

Project 3 — VoxOps

Real-Time Voice AI Helpdesk Agent with MCP and AgentOps

Main Idea

A real-time voice agent for an IT helpdesk or internal support department. A user speaks through the browser; the agent converts speech to text, identifies intent, retrieves relevant support information, calls approved tools, completes an action, responds using generated speech, and transfers to a human when uncertain. Current Voice AI roles emphasize real-time STT/TTS, low latency, agent workflows, production reliability, observability, and evaluation.

Example Interaction

- User: “My laptop cannot connect to the university Wi-Fi.”
- Agent: “I found two common causes. Before creating a ticket, may I check whether your university account is active?”
- User: “Yes.”
- Agent calls `check_account_status`, confirms the account is active, and offers to create a network-support ticket and schedule a callback
- Agent calls `create_ticket` and `schedule_callback` once the user confirms

Recommended Architecture

Browser microphone → voice activity detection → streaming speech-to-text → LangGraph voice orchestrator → intent and confidence detection → routed to knowledge search / MCP tools / human handoff → text-to-speech → browser audio response.

Technology Stack

Layer	Tool / Technology
Speech-to-text	faster-whisper
Voice activity detection	Silero VAD
Text-to-speech	Piper or Kokoro
Orchestration	LangGraph
Tool interface	Model Context Protocol (MCP)
Backend	FastAPI with WebSockets
State / cache	Redis
Database	PostgreSQL or Supabase
Observability	Langfuse
Deployment	Docker on GCP, GitHub Actions CI/CD

Scope Constraint

Limit the domain to IT helpdesk only, with exactly four tools: `search_support_knowledge`, `check_ticket_status`, `create_support_ticket`, and `schedule_callback`. Do not attempt to integrate ten different business systems during the three-week build.

Key Feature Areas

Tool reliability — every tool should have:

- Pydantic input validation and structured JSON output
- Timeout, retry policy, and idempotency key
- Error response, permission requirement, and audit log

Human-in-the-loop

Require confirmation before: creating a ticket, changing an appointment, sending information, or closing a request.

Transfer to a human when: confidence is low, speech cannot be understood, a tool fails repeatedly, or the user explicitly requests a person.

Conversation handling

- Streaming responses and interruption/barge-in support
- Conversation memory, silence detection, repeated-question handling
- Fallback messages and recovery after a failed tool call

Day-Wise Plan

Week 1 — Basic Real-Time Voice Loop

Day	Tasks
Day 1	• Build the browser microphone interface (audio capture in-browser)
Day 2	• Integrate faster-whisper for speech-to-text • Test transcription on sample audio
Day 3	• Add Silero VAD for voice-activity detection
Day 4	• Set up the FastAPI WebSocket server • Stream audio from browser to backend
Day 5	• Add text-to-speech using Piper or Kokoro
Day 6	• Wire the full loop: mic → VAD → STT → basic response → TTS → playback
Day 7	• Measure transcription and end-to-end response latency • Buffer day for fixes

Week 2 — Agent Workflow & MCP

Day	Tasks
Day 1	Design the LangGraph state machine and intent-routing logic
Day 2	Implement the search_support_knowledge MCP tool
Day 3	Implement check_ticket_status and create_support_ticket MCP tools
Day 4	- Implement schedule_callback MCP tool - Add retries, timeouts, and idempotency keys to all tools
Day 5	- Add the user-confirmation flow before any state-changing action - Set up the ticket database schema
Day 6	Implement human handoff logic (low confidence, repeated tool failure, explicit request)
Day 7	- Add conversation memory using Redis - Buffer day for integration testing

Week 3 — Evaluation & Production Deployment

Day	Tasks
Day 1	Record a set of test conversations: different accents, background noise, ambiguous requests
Day 2	Implement word error rate (WER) and intent-accuracy scoring
Day 3	Implement tool-selection/argument accuracy and task-completion-rate scoring
Day 4	- Implement handoff-accuracy, end-to-end latency, and failure-recovery metrics - Wire up Langfuse tracing and a latency dashboard
Day 5	- Dockerize the app - Set up CI/CD with GitHub Actions
Day 6	- Deploy to GCP - Publish the public browser demo
Day 7	- Draw the architecture diagram - Record the demo video - Final polish and review

CV Positioning

VoxOps demonstrates real-time Voice AI, streaming systems, STT/TTS integration, agent tool calling under latency constraints, reliability engineering, human handoff, and cloud deployment — a genuinely different skill story from both NOVA (research RAG) and MemoraGraph (memory infrastructure).

